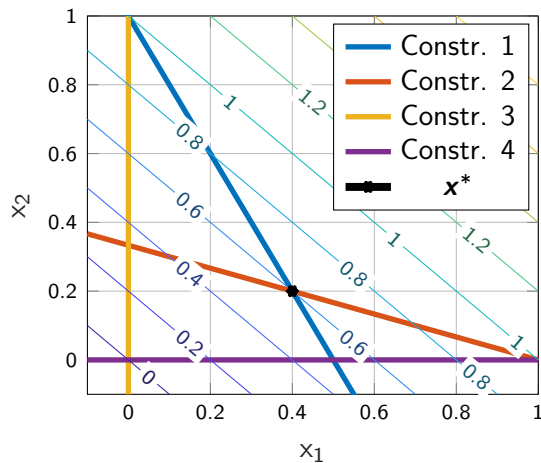


Solution of Exercise 1

Exercise 1.1 Optimal points

$$f(\mathbf{x}) = x_1 + x_2 \rightarrow \mathbf{x}^* = [0.4, 0.2]^T, f(\mathbf{x}^*) = 0.6$$



$$f(\mathbf{x}) = -x_1 - x_2 \rightarrow \text{Unbounded solution}$$

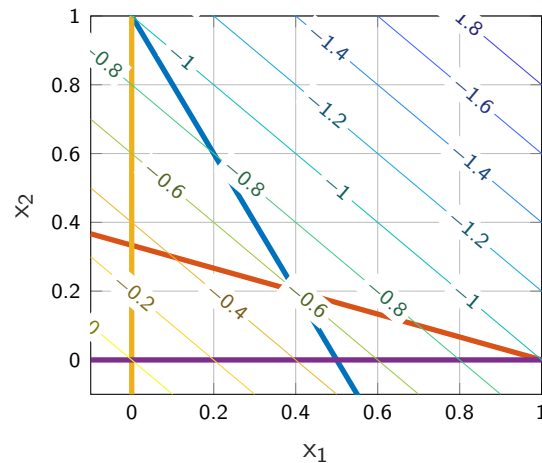
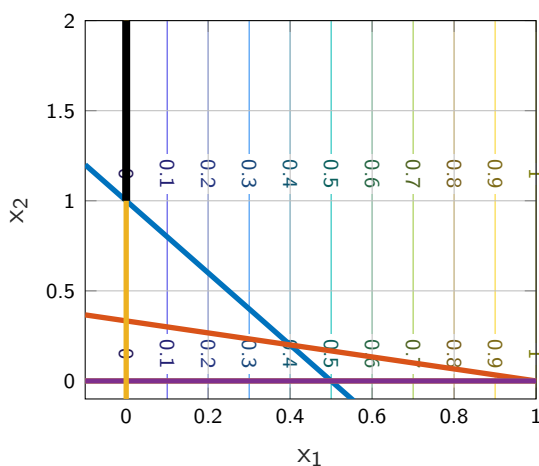


Figure 1: Contour lines and optimal points for the different cost functions.

Add-On:

$$f(\mathbf{x}) = x_1 \rightarrow \mathbf{x}^* = [0, x_2]^T \forall x_2 \geq 1, f(\mathbf{x}^*) = 0$$



$$f(\mathbf{x}) = \max(x_1, x_2) \rightarrow \mathbf{x}^* = [\frac{1}{3}, \frac{1}{3}]^T, f(\mathbf{x}^*) = \frac{1}{3}$$

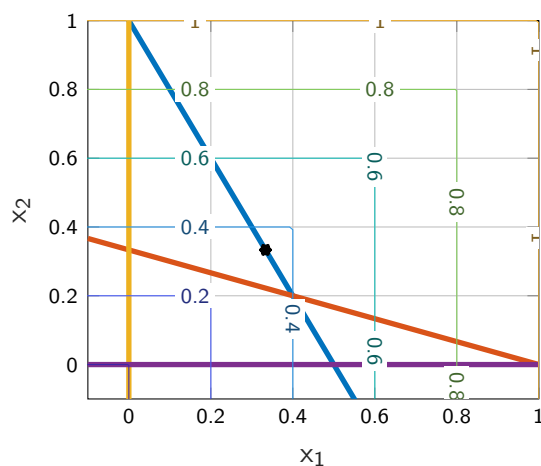


Figure 2: Contour lines and optimal points for the different cost functions.

Exercise 1.2 Convexity

Using Theorem 2.2 (Definiteness) and Theorem 2.3 from the script.

$$(1) f(\mathbf{x}) = \frac{x_1^2}{x_2}, \quad x_1, x_2 > 0$$

Gradient and Hessian matrix:

$$\nabla f(\mathbf{x}) = \begin{bmatrix} \frac{2x_1}{x_2} \\ -\frac{x_1^2}{x_2^2} \end{bmatrix}$$

$$\nabla^2 f(\mathbf{x}) = \begin{bmatrix} \frac{2}{x_2} & -\frac{2x_1}{x_2^2} \\ -\frac{2x_1}{x_2^2} & \frac{2x_1^2}{x_2^3} \end{bmatrix}$$

Examination of the definiteness of the Hessian according to Theorem 2.2:

$$\begin{aligned} \mathbf{p}^\top \nabla^2 f(\mathbf{x}) \mathbf{p} &= p_1^2 \frac{2}{x_2} - \frac{4x_1}{x_2^2} p_1 p_2 + \frac{2x_1^2}{x_2^3} p_2^2 \\ &= \frac{2}{x_2} (p_1^2 - 2 \frac{x_1}{x_2} p_1 p_2 + \frac{x_1^2}{x_2^2} p_2^2) \\ &= \frac{2}{x_2} (p_1 - \frac{x_1}{x_2} p_2)^2 \geq 0, \quad \forall x_1, x_2 > 0 \end{aligned}$$

$\Rightarrow \nabla^2 f(\mathbf{x})$ is positive semidefinite $\Rightarrow f$ is convex on its domain of definition.

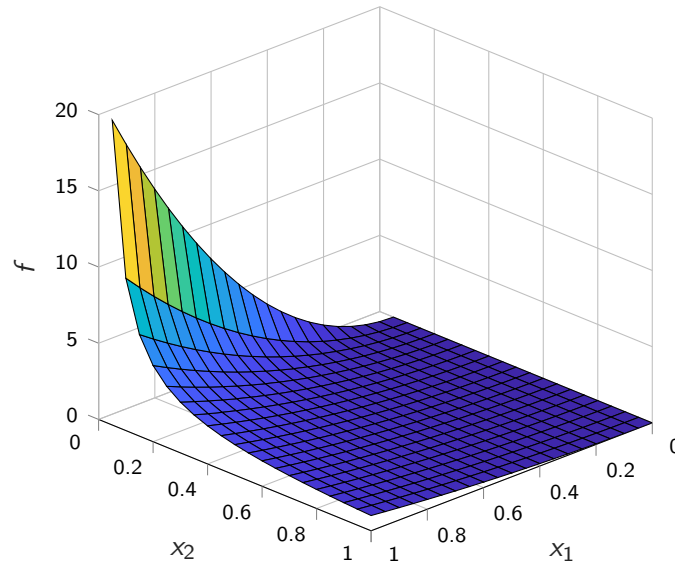


Figure 3: Contour line plot of function (1).

$$(2) f(\mathbf{x}) = \frac{1}{2\pi} \exp\left(-\frac{x_1^2 + x_2^2}{2}\right), \quad \mathbf{x} \in \mathbb{R}^2$$

Gradient and Hessian matrix:

$$\nabla f(\mathbf{x}) = \begin{bmatrix} \frac{1}{2\pi} \exp\left(-\frac{x_1^2 + x_2^2}{2}\right) (-x_1) \\ \frac{1}{2\pi} \exp\left(-\frac{x_1^2 + x_2^2}{2}\right) (-x_2) \end{bmatrix}$$

$$\nabla^2 f(\mathbf{x}) = \begin{bmatrix} \frac{1}{2\pi} \exp\left(-\frac{x_1^2 + x_2^2}{2}\right) (x_1^2 - 1) & \frac{1}{2\pi} \exp\left(-\frac{x_1^2 + x_2^2}{2}\right) x_1 x_2 \\ \frac{1}{2\pi} \exp\left(-\frac{x_1^2 + x_2^2}{2}\right) x_1 x_2 & \frac{1}{2\pi} \exp\left(-\frac{x_1^2 + x_2^2}{2}\right) (x_2^2 - 1) \end{bmatrix}$$

Examination of the definiteness of the Hessian according to Theorem 2.2:

$$\begin{aligned} \mathbf{p}^T \nabla^2 f(\mathbf{x}) \mathbf{p} &= p_1^2 \frac{1}{2\pi} \exp\left(-\frac{x_1^2 + x_2^2}{2}\right) (x_1^2 - 1) \\ &\quad + p_1 p_2 \frac{1}{2\pi} \exp\left(-\frac{x_1^2 + x_2^2}{2}\right) 2x_1 x_2 \\ &\quad + p_2^2 \frac{1}{2\pi} \exp\left(-\frac{x_1^2 + x_2^2}{2}\right) (x_2^2 - 1) \\ &= \frac{1}{2\pi} \exp\left(-\frac{x_1^2 + x_2^2}{2}\right) (p_1^2 (x_1^2 - 1) + 2p_1 p_2 x_1 x_2 + p_2^2 (x_2^2 - 1)) \\ &= \frac{1}{2\pi} \exp\left(-\frac{x_1^2 + x_2^2}{2}\right) ((p_1 x_1 + p_2 x_2)^2 - p_1^2 - p_2^2) \end{aligned}$$

For $x_1, x_2 = 0 \Rightarrow \mathbf{p}^T \nabla^2 f(\mathbf{x}) \mathbf{p} < 0 \Rightarrow f$ is strictly concave.

For $x_1, x_2 = 1 \Rightarrow \mathbf{p}^T \nabla^2 f(\mathbf{x}) \mathbf{p} = 2p_1 p_2 \Rightarrow$ undefined.

$\Rightarrow f$ is neither concave nor convex on its domain of definition.

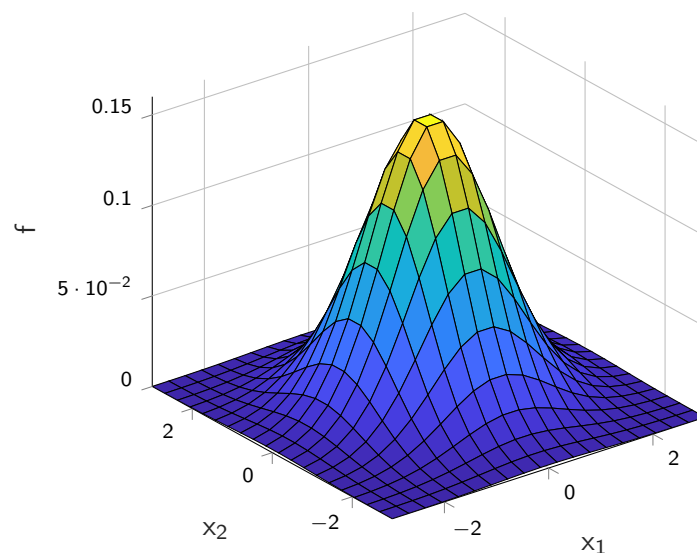


Figure 4: Contour line plot of function (2).

Exercise 1.3 Line search**Gradient method:** $\mathbf{x}^{k+1} = \mathbf{x}^k + \alpha^k \mathbf{s}^k$, $\mathbf{s}^k = -\nabla f(\mathbf{x}^k)$

- Gradient

$$\nabla f(\mathbf{x}) = \begin{bmatrix} 2(x_1 - 1) + x_2 \\ 4(2x_2 - 4) + x_1 \end{bmatrix}$$

- 1. step: $\mathbf{x}^0 = [-1, 1]^T$

$$\nabla f(\mathbf{x}^0) = \begin{bmatrix} -3 \\ -9 \end{bmatrix} \Rightarrow \mathbf{s}^0 = -\nabla f(\mathbf{x}^0) = \begin{bmatrix} 3 \\ 9 \end{bmatrix}$$

Determining the step size α : $\min_{\alpha \in \mathbb{R}} f(\mathbf{x} + \alpha \mathbf{s})$

$$f(\mathbf{x} + \alpha \mathbf{s}) = (x_1 + s_1 \alpha - 1)^2 + (x_1 + s_1 \alpha)(x_2 + s_2 \alpha) + (2(x_2 + s_2 \alpha) - 4)^2$$

$$\begin{aligned} \frac{\partial f(\mathbf{x} + \alpha \mathbf{s})}{\partial \alpha} &= 2(x_1 + \alpha s_1 - 1)s_1 + s_1(x_2 + \alpha s_2) + s_2(x_1 + \alpha s_1) + 2(2(x_2 + \alpha s_2) - 4)s_2 \\ &= (2s_1^2 + s_1 s_2 + s_1 s_2 + 8s_2^2)\alpha + 2s_1(x_1 - 1) + s_1 x_2 + s_2 x_1 + 4s_2(2x_2 - 4) \stackrel{!}{=} 0 \end{aligned}$$

$$\Rightarrow \alpha = \frac{-2s_1(x_1 - 1) - s_1 x_2 - s_2 x_1 - 4s_2(2x_2 - 4)}{2s_1^2 + 2s_1 s_2 + 8s_2^2} \quad (1)$$

Insert values for first step:

$$\alpha^0 = \frac{-6(-1 - 1) - 3 + 9 - 36(2 - 4)}{18 + 54 + 648} = \frac{1}{8}$$

$$\Rightarrow \mathbf{x}^1 = [-1, 1]^T + \frac{1}{8}[3, 9]^T = [-\frac{5}{8}, \frac{17}{8}]^T$$

- 2. step:

$$\nabla f(\mathbf{x}^1) = \begin{bmatrix} -\frac{9}{8} \\ \frac{3}{8} \end{bmatrix} \Rightarrow \mathbf{s}^1 = -\nabla f(\mathbf{x}^1) = \begin{bmatrix} \frac{9}{8} \\ -\frac{3}{8} \end{bmatrix}$$

Determine step size α^1 from (1): $\alpha^1 = \frac{1}{2}$

$$\Rightarrow \mathbf{x}^2 = [-\frac{5}{8}, \frac{17}{8}]^T + \frac{1}{2}[\frac{9}{8}, -\frac{3}{8}]^T = [-\frac{1}{16}, \frac{31}{16}]^T$$

Newton's method: $\mathbf{s}^k = -[\nabla^2 f(\mathbf{x}^k)]^{-1} \nabla f(\mathbf{x}^k)$

- Hessian matrix

$$\nabla^2 f(\mathbf{x}) = \begin{bmatrix} 2 & 1 \\ 1 & 8 \end{bmatrix} \Rightarrow [\nabla^2 f(\mathbf{x})]^{-1} = \frac{1}{15} \begin{bmatrix} 8 & -1 \\ -1 & 2 \end{bmatrix}$$

- 1. step: $\nabla f(\mathbf{x}^0) = \begin{bmatrix} -3 \\ -9 \end{bmatrix}$

$$\mathbf{s}^0 = -\frac{1}{15} \begin{bmatrix} 8 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} -3 \\ -9 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\text{Step size } \alpha = 1: \mathbf{x}^1 = \mathbf{x}^0 + \mathbf{s}^0 = [-1, 1]^T + [1, 1]^T = [0, 2]^T = \mathbf{x}^*$$

Add-On: Conjugate gradient method:

$$\mathbf{s}^0 = -\nabla f(\mathbf{x}^0)$$

$$\mathbf{s}^k = -\nabla f(\mathbf{x}^k) + \beta^k \mathbf{s}^{k-1} \quad \text{with} \quad \beta^k = \frac{\|\nabla f(\mathbf{x}^k)\|^2}{\|\nabla f(\mathbf{x}^{k-1})\|^2}$$

- **1. step:** Identical to gradient method: $\mathbf{x}^1 = [-\frac{5}{8}, \frac{17}{8}]^T$

- **2. step:** $\nabla f(\mathbf{x}^1) = \begin{bmatrix} -\frac{9}{8} \\ \frac{3}{8} \end{bmatrix}$

$$\Rightarrow \beta^1 = \frac{\|\nabla f(\mathbf{x}^1)\|^2}{\|\nabla f(\mathbf{x}^0)\|^2} = \frac{\frac{81}{64} + \frac{9}{64}}{9 + 81} = \frac{1}{64}$$

$$\Rightarrow \mathbf{s}^1 = \begin{bmatrix} \frac{9}{8} \\ -\frac{3}{8} \end{bmatrix} + \frac{1}{64} \begin{bmatrix} 3 \\ 9 \end{bmatrix} = \begin{bmatrix} \frac{75}{64} \\ -\frac{15}{64} \end{bmatrix}$$

Determine step size α^1 from (1): $\alpha^1 = \frac{8}{15}$

$$\Rightarrow \mathbf{x}^2 = [-\frac{5}{8}, \frac{17}{8}]^T + \frac{8}{15} [\frac{75}{64}, -\frac{15}{64}]^T = [0, 2]^T = \mathbf{x}^*$$